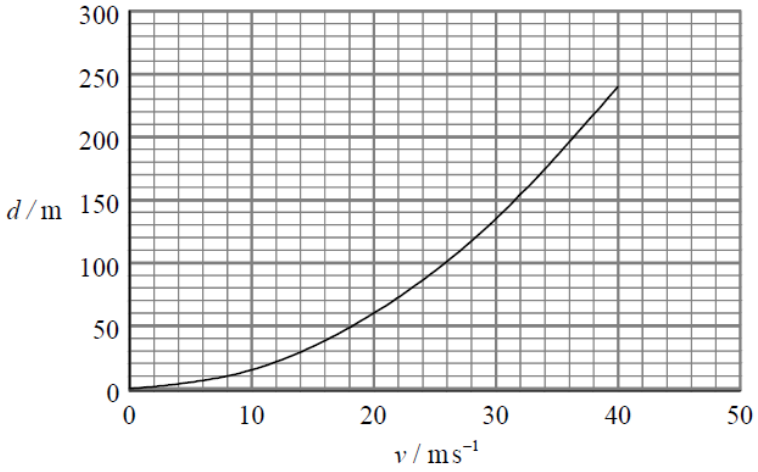
Mechanics Worksheet

1.

This question is about the braking distance of a car and the power developed by the car engine.

The **minimum braking distance** is the shortest distance a car travels without skidding from the moment the brakes are applied until the moment the car comes to rest

The graph below shows how the minimum braking distance d varies with the initial speed v of a car travelling along a straight, horizontal road.



(a) By choosing **two** data points show that the graph suggests that the minimum braking distance depends on the square of the initial speed. *[4]*

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(b) Explain why, if the braking force is constant, you would expect theoretically the braking distance to depend on the square of the speed. *[3]*

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(c) The car has a mass of 1500 kg and is moving with an initial speed of $20\,\mathrm{m\,s^{-1}}$ when the brakes are applied. Calculate

(i) the time that it takes the car to come to rest. *[3]*

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(ii) the average braking force acting on the car. *[3]*

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(d) The **stopping distance** of a car is the distance that the car travels in coming to rest from the moment that the driver decides to apply the brakes. The **stopping distance** is greater than the **minimum braking distance**. Explain why you think that this is so. *[2]*

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(e) Using the same axes as for the graph given at the start of the question sketch a graph to show how the stopping distance varies with initial speed. Explain the shape of the graph that you have drawn. *[3]*

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(f) If the car were travelling down an incline would you expect the minimum braking distance for a given initial speed to be greater than or less than the braking distance on the level for the same road conditions? Explain. *[2]*

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The next part of the question looks at the power developed by the car engine.

(g) Travelling at a constant speed of $40\ \mathrm{m\,s^{-1}}$ the car engine uses 1 litre of fuel for every 12 km travelled. At what rate does the engine use fuel? *[2]*

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(h) The combustion of 1 litre of fuel releases 35 MJ of energy and at a speed of $40\ \mathrm{m\,s^{-1}}$ the car engine is 25 % efficient. Determine the power output of the car engine at this speed? *[2]*

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(i) Use your answer to (h) above to estimate the average drag force acting on the car when travelling at a constant speed of $40\ \mathrm{m\,s^{-1}}$. *[1]*

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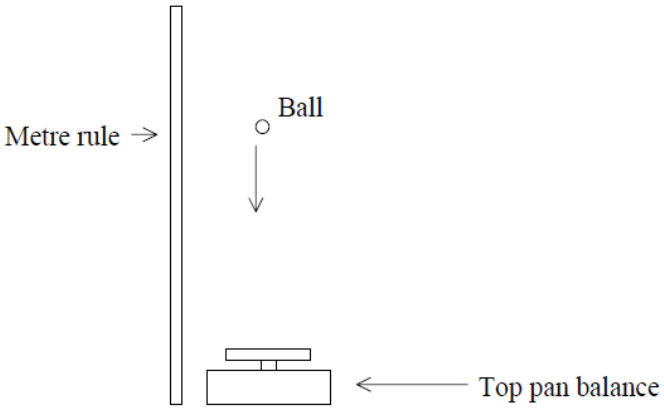
2.

This question is about a bouncing ball and contact time.



Miguel has devised a method to measure how long a bouncing ball is in contact with the surface from which it bounces. The method consists of dropping the ball on to the scale pan of a top pan balance as shown in the diagram above. The balance is calibrated in newtons and Miguel records the maximum reading on the scale, the height from which the ball is dropped and the height to which it bounces.

Miguel obtains the following information.

Height from which the ball is dropped	=	0.80 m
Height to which the ball bounces	=	0.60 m
Maximum reading on the balance scale	=	50.0 N

The mass of the ball is 0.20 kg and the acceleration due to gravity is taken to have a value of 10 $m\,s^{-2}$.

(a) Calculate

(i) the speed of the ball when it strikes the scale pan. *[1]*

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(ii) the speed of the ball when it leaves the scale pan. *[1]*

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(iii) the total change in momentum of the ball between striking and leaving the scale pan. *[2]*

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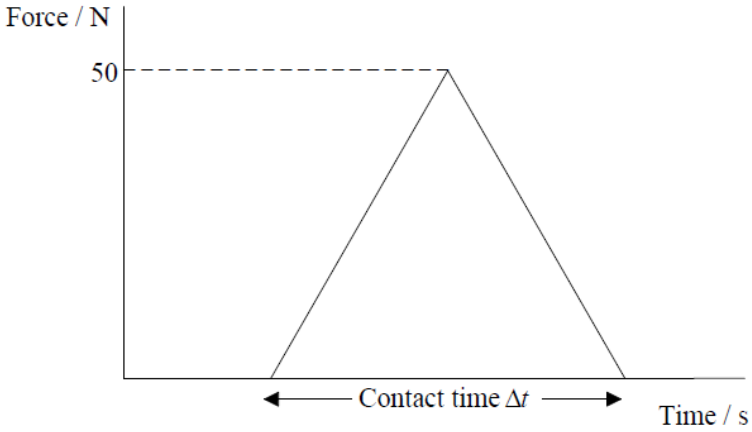
(b) Miguel assumes that the contact force between the ball and the scale pan varies with time as shown below.



(i) What does the area under the graph represent? *[1]*

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(ii) Calculate the contact time Δt. *[2]*

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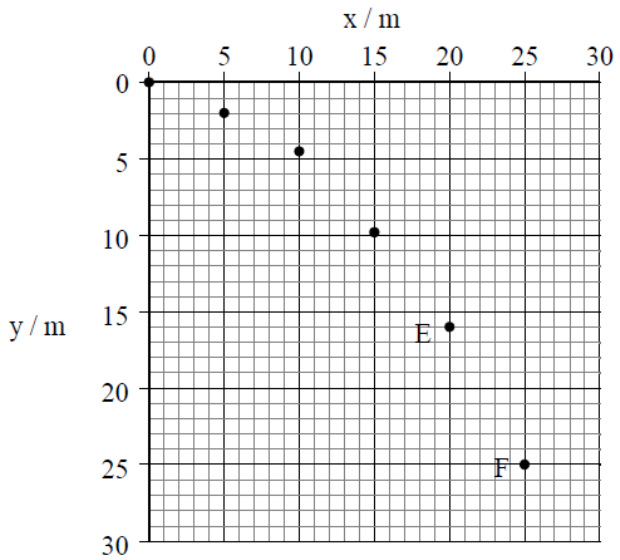
(c) Miguel now drops another ball from the same height on to the scale pan. This ball is the same mass as the first ball but is made of a harder material. Sketch, using the same axes as in (b) above, the shape of the graph Miguel might expect to get for this ball. *[2]*

3.

Projectile motion on a planet

A projectile is launched horizontally from a cliff on a planet in a distant solar system. The graph below plots the horizontal (x) and vertical (y) positions of the projectile **every 0.5 seconds**.



(a) Determine the initial velocity with which the projectile was launched. *[2]*

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(b) How can you tell from the plotted data that the planet's atmosphere had no significant effect on the motion of the projectile? *[2]*

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(c) State **two** reasons why the value of the acceleration due to gravity on this or any other planet is likely to be different from that on Earth. *[2]*

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(d) Draw a vector on the graph to represent the **displacement** of the projectile between points E and F of the motion. Then draw vectors to represent the horizontal and vertical **components** of this displacement. *[3]*

(e) Determine the **vertical** component of the average velocity of the projectile between points E and F. *[2]*

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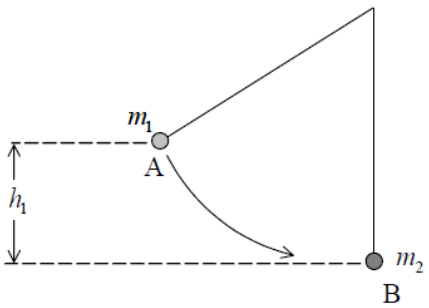
(f) Another projectile is fired at **half the speed** of the first one. On the graph opposite, plot the positions of this projectile for time intervals of 0.5 s. *[2]*

4.

Part 2 Pendulum collision

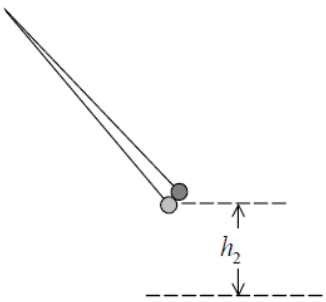
Two balls A and B, of masses m_1 and m_2 respectively, are suspended from a common point by strings of equal length. Ball A is pulled aside to the left, rising a height h_1, as shown in diagram 1 and is then released.

Diagram 1



Ball A swings down, **sticks** to ball B, and the two balls **together** swing up to the right to a height h_2 as shown in diagram 2.

Diagram 2



(a) Deduce an expression for

(i) the speed of m_1 immediately before it collides with m_2. *[2]*

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(ii) the speed of m_1 and m_2 immediately after collision. *[4]*

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(b) If the expression for the speed of m_1 and m_2 immediately after collision is known, state the name of the principle (law) of physics that enable an expression for the height h_2 to be found in terms of h_1, m_1, m_2 and g. *[1]*

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(c) Explain why the height h_2 will always be less than the height h_1. *[1]*

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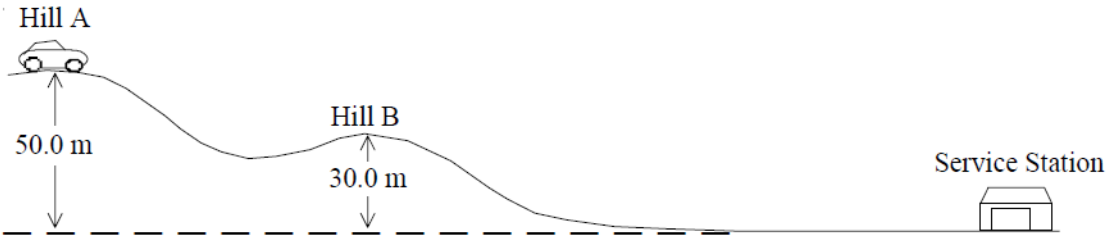
5.

Part 2 A car rolling down a hill

A car runs out of fuel at the top of hill A. The driver hopes to get to the service station by letting the car roll downhill with its engine switched off. There is a small hill, B, in between as shown. The distance from hill A to hill B along the road is 0.2 km and from hill B to the service station is 0.4 km.



The top of hill A is 50.0 m above the service station and the top of hill B is 30.0 m above the service station.

(a) When the car reaches hill B its speed is $5.0\,\mathrm{m\,s^{-1}}$. Assuming that the car starts from rest at hill A, show that the average frictional force acting on the car is 750 N. The mass of the car is 800 kg. *[5]*

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(b) Assuming that the frictional force remains constant throughout, determine whether or not the car reaches the service station. *[3]*

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(c) Besides friction in the wheels and tyres, name another source of frictional resistance for the moving car, and explain why it will not in fact remain constant during the motion. *[2]*

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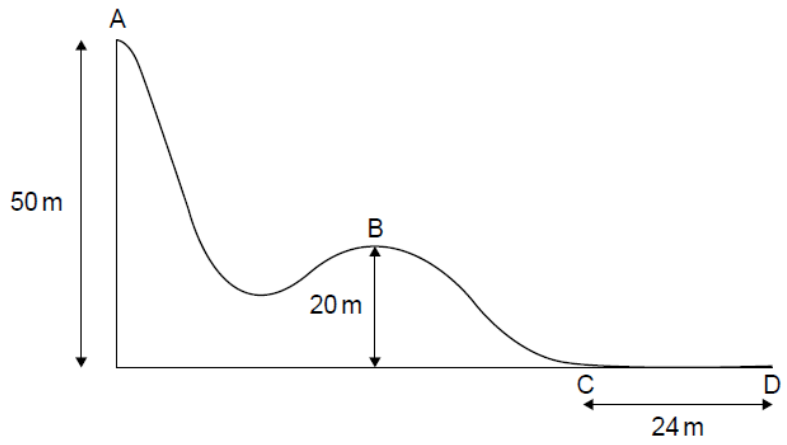
6.

1. The diagram below shows part of a downhill ski course which starts at point A, 50 m above level ground. Point B is 20 m above level ground.



(a) A skier of mass 65 kg starts from rest at point A and during the ski course some of the gravitational potential energy transferred to kinetic energy.

(i) From A to B, 24 % of the gravitational potential energy transferred to kinetic energy. Show that the velocity at B is $12\,m\,s^{-1}$. [2]

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(ii) Some of the gravitational potential energy transferred into internal energy of the skis, slightly increasing their temperature. Distinguish between internal energy and temperature. [2]

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(b) (i) The dot on the following diagram represents the skier as she passes point B. Draw and label the vertical forces acting on the skier. [2]

(ii) The hill at point B has a circular shape with a radius of 20 m. Determine whether the skier will lose contact with the ground at point B. [3]

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(c) The skier reaches point C with a speed of $8.2\,\mathrm{m\,s^{-1}}$. She stops after a distance of 24 m at point D.

Determine the coefficient of dynamic friction between the base of the skis and the snow. Assume that the frictional force is constant and that air resistance can be neglected. [3]

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(d) At the side of the course flexible safety nets are used. Another skier of mass 76 kg falls normally into the safety net with speed $9.6\,m\,s^{-1}$.

(i) Calculate the impulse required from the net to stop the skier and state an appropriate unit for your answer. [2]

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(ii) Explain, with reference to change in momentum, why a flexible safety net is less likely to harm the skier than a rigid barrier. [2]

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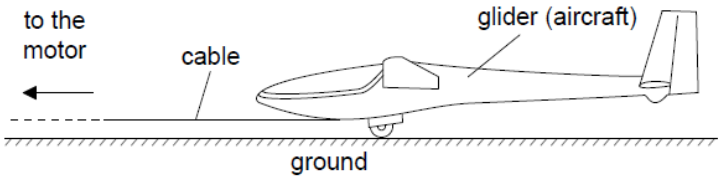
7.

1. A glider is an aircraft with no engine. To be launched, a glider is uniformly accelerated from rest by a cable pulled by a motor that exerts a horizontal force on the glider throughout the launch.



(a) The glider reaches its launch speed of $27.0\,\mathrm{m\,s^{-1}}$ after accelerating for 11.0 s. Assume that the glider moves horizontally until it leaves the ground. Calculate the total distance travelled by the glider before it leaves the ground. [2]

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(b) The glider and pilot have a total mass of 492 kg. During the acceleration the glider is subject to an average resistive force of 160 N. Determine the average tension in the cable as the glider accelerates. [3]

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(c) The cable is pulled by an electric motor. The motor has an overall efficiency of 23 %. Determine the average power input to the motor. [3]

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(d) The cable is wound onto a cylinder of diameter 1.2 m. Calculate the angular velocity of the cylinder at the instant when the glider has a speed of 27 m s^{-1}. Include an appropriate unit for your answer. [2]

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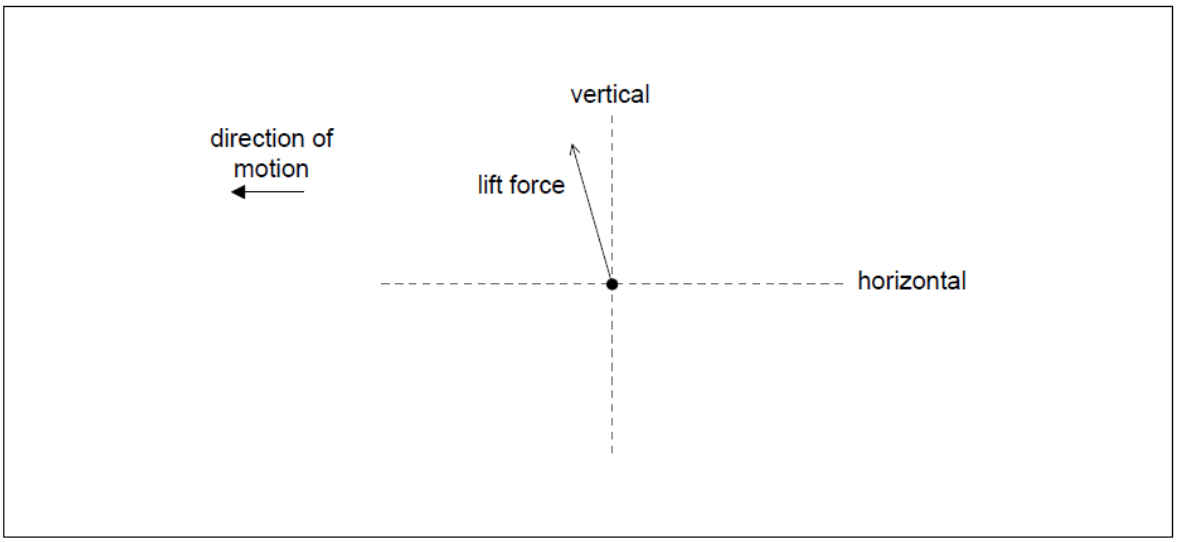
(e) After takeoff the cable is released and the unpowered glider moves horizontally at constant speed. The wings of the glider provide a lift force. The diagram shows the lift force acting on the glider and the direction of motion of the glider.



Draw the forces acting on the glider to complete the free-body diagram. The dotted lines show the horizontal and vertical directions. [2]

(f) Explain, using appropriate laws of motion, how the forces acting on the glider maintain it in level flight. [2]

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(g) At a particular instant in the flight the glider is losing 1.00 m of vertical height for every 6.00 m that it goes forward horizontally. At this instant, the horizontal speed of the glider is 12.5 m s^{-1}. Calculate the **velocity** of the glider. Give your answer to an appropriate number of significant figures. [3]

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